

Exp 10	Digital Speech Processing and Noise Reduction
Date:	10.1. Implementation of Spectral Subtraction for Speech Noise Reduction

Code:

```

clc;

clear;

close all;

%% Input Parameters

Fs = 8000;           % Sampling Frequency
t = 0:1/Fs:3;        % Time Vector

%% Signals

x = sin(2*pi*200*t) + 0.5*sin(2*pi*400*t); % Clean signal
noise = 0.3*randn(size(x));           % Noise
xn = x + noise;                       % Noisy signal

%% FFT-based Spectral Subtraction

Y = fft(xn);           % FFT of noisy signal
N = fft(noise);         % FFT of noise estimate
MagY = abs(Y);          % Magnitude of noisy signal
MagN = abs(N);          % Magnitude of noise
PhaseY = angle(Y);      % Phase of noisy signal

% Spectral subtraction (ensure non-negative magnitudes)
MagE = max(MagY - MagN, 0);

% Reconstruct enhanced signal
xe = real(ifft(MagE .* exp(1j*PhaseY)));

```

```
%% --- Time Domain Plots ---
```

```
figure;
```

```
subplot(3,1,1), plot(t, x), title('Clean Speech'), xlabel('Time (s)'), ylabel('Amplitude'), grid on
```

```
subplot(3,1,2), plot(t, xn), title('Noisy Speech'), xlabel('Time (s)'), ylabel('Amplitude'), grid on
```

```
subplot(3,1,3), plot(t, xe), title('Enhanced Speech'), xlabel('Time (s)'), ylabel('Amplitude'), grid on
```

```
%% --- Spectrograms ---
```

```
figure;
```

```
subplot(3,1,1), spectrogram(x, 256, 200, 256, Fs, 'yaxis'), title('Clean Spectrogram')
```

```
subplot(3,1,2), spectrogram(xn, 256, 200, 256, Fs, 'yaxis'), title('Noisy Spectrogram')
```

```
subplot(3,1,3), spectrogram(xe, 256, 200, 256, Fs, 'yaxis'), title('Enhanced Spectrogram')
```

```
%% --- Magnitude Spectra ---
```

```
figure;
```

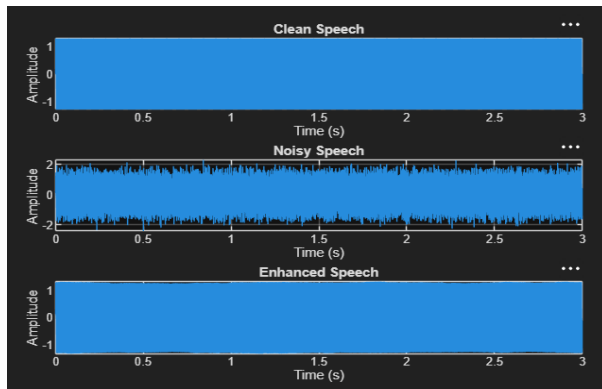
```
subplot(3,1,1), plot(abs(fft(x))), title('Clean Spectrum'), xlabel('Frequency Bin'), ylabel('Magnitude'),  
grid on
```

```
subplot(3,1,2), plot(abs(fft(xn))), title('Noisy Spectrum'), xlabel('Frequency Bin'), ylabel('Magnitude'),  
grid on
```

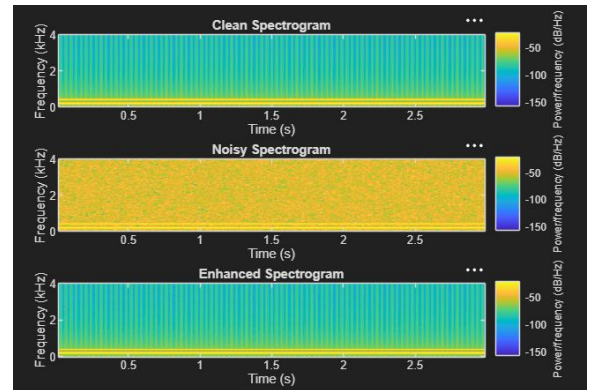
```
subplot(3,1,3), plot(abs(fft(xe))), title('Enhanced Spectrum'), xlabel('Frequency Bin'),  
ylabel('Magnitude'), grid on
```

Output:

Objective 1:



Objective 2:



Objective 3:

